

1 Summary

Model (Run-name): model3 (test)

Iteration: 4

TAC: 62209.252067288704

Took: 28.41644024848938s

Total Time: 50.66274952888489s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

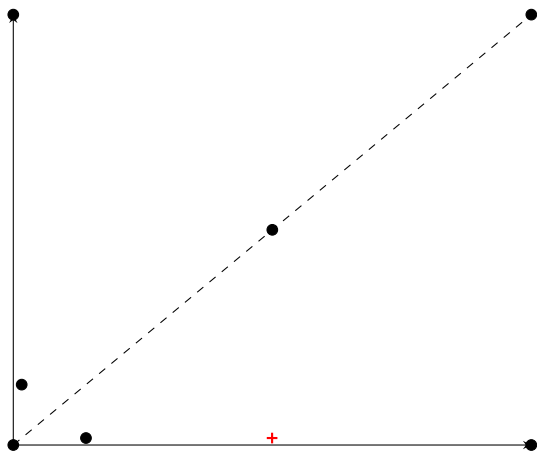
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,4,2}$	1.0	$dt_{3,1,1}$	66.7	$q_{1,4,2}$	586.96
$z_{2,2,1}$	1.0	$dt_{3,1,2}$	46.84594594594559	$q_{2,2,1}$	647.2099999999999
$z_{2,5,1}$	1.0	$dt_{3,5,1}$	22.3	$q_{2,5,1}$	530.45
$z_{3,1,1}$	1.0	$dt_{3,5,2}$	13.545945945945633	$q_{3,1,1}$	760.0
$z_{3,5,1}$	1.0	$dt_{4,3,1}$	50.0	$q_{3,5,1}$	1013.8399999999987
$z_{4,3,1}$	1.0	$dt_{4,3,2}$	111.1	$q_{4,3,1}$	1539.72
$z_{5,4,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{5,4,1}$	1047.8899999999935
$z_{cu,3}$	1.0	$dt_{5,4,2}$	23.568851441820925	$q_{cu,3}$	610.4399999999984
$z_{cu,5}$	1.0	$dt_{cu,3}$	66.84594594594486	$q_{cu,5}$	1311.5200000000063
$A_{1,4,2}^\beta$	5.9882685928326005	$dt_{cu,5}$	99.69717514124329	$t_{1,1}^{(H)}$	160.0
$A_{2,2,1}^\beta$	7.4352676726747875	$f_{1,4,2}^H$	8.8	$t_{1,2}^{(H)}$	160.0
$A_{2,5,1}^\beta$	0.5579246324326862	$f_{2,2,1}^H$	5.825472547254917	$t_{1,3}^{(H)}$	93.3
$A_{3,1,1}^\beta$	2.2768807044848223	$f_{2,5,1}^H$	4.77452745274533	$t_{2,1}^{(H)}$	248.9
$A_{3,5,1}^\beta$	6.733274165260775	$f_{3,1,1}^H$	4.717566728739804	$t_{2,2}^{(H)}$	137.8
$A_{4,3,1}^\beta$	6.40634053317395	$f_{3,5,1}^H$	10.082433271260093	$t_{2,3}^{(H)}$	137.8
$A_{5,4,1}^\beta$	7.341424653538622	$f_{4,3,1}^H$	12.6	$t_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	4.764278257928811	$f_{5,4,1}^H$	7.861140285071216	$t_{3,2}^{(H)}$	106.84594594594522
$A_{cu,5}^\beta$	6.370670015678422	$f_{1,4,2}^C$	6.211216931216901	$t_{3,3}^{(H)}$	106.84594594594486
$A_{1,4,2}$	26.128323085758318	$f_{2,2,1}^C$	6.1	$t_{4,1}^{(H)}$	271.1
$A_{2,2,1}$	30.93278282491032	$f_{2,5,1}^C$	4.7745274527453265	$t_{4,2}^{(H)}$	148.9
$A_{2,5,1}$	4.010660819597746	$f_{3,1,1}^C$	7.6	$t_{4,3}^{(H)}$	148.9
$A_{3,1,1}$	13.737197536033536	$f_{3,5,1}^C$	9.125472547254674	$t_{5,1}^{(H)}$	198.9
$A_{3,5,1}$	30.750913778350093	$f_{4,3,1}^C$	8.4	$t_{5,2}^{(H)}$	139.6971751412433
$A_{4,3,1}$	23.670439786681058	$f_{5,4,1}^C$	14.19127082289588	$t_{5,3}^{(H)}$	139.6971751412433
$A_{5,4,1}$	38.682379459353704	$RecLMTD_{1,4,2}$	0.03783745846888118	$t_{1,1}^{(C)}$	160.0
$A_{cu,3}$	13.6435791010403	$RecLMTD_{2,2,1}$	0.04062493688474249	$t_{1,2}^{(C)}$	60.0
$A_{cu,5}$	22.410514202757604	$RecLMTD_{2,5,1}$	0.02833130321344378	$t_{1,3}^{(C)}$	60.0
$dt_{1,4,2}$	43.87167630057803	$RecLMTD_{3,1,1}$	0.015363970928458967	$t_{2,1}^{(C)}$	221.7
$dt_{1,4,3}$	11.1	$RecLMTD_{3,5,1}$	0.05127487499860754	$t_{2,2}^{(C)}$	115.6
$dt_{2,2,1}$	27.2	$RecLMTD_{4,3,1}$	0.013067228988828773	$t_{2,3}^{(C)}$	115.6
$dt_{2,2,2}$	22.2	$RecLMTD_{5,4,1}$	0.043708528416562	$t_{3,1}^{(C)}$	221.1
$dt_{2,5,1}$	44.5	$RecLMTD_{cu,3}$	0.018997841288061607	$t_{3,2}^{(C)}$	37.8
$dt_{2,5,2}$	12.570732488352636	$RecLMTD_{cu,5}$	0.015158682618526556	$t_{3,3}^{(C)}$	37.8

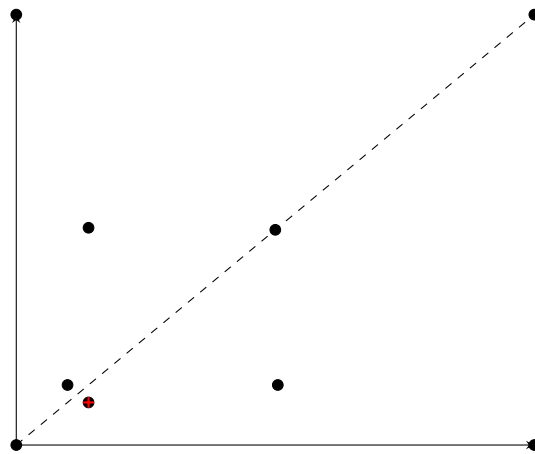
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,1}^{H,out}$	515.6908027006716
$t_{4,2}^{(C)}$	116.12832369942198	$b_{1,4,2}^{C,in}$	510.5620317460293
$t_{4,3}^{(C)}$	82.2	$b_{2,2,1}^{C,in}$	705.16
$t_{5,1}^{(C)}$	204.4	$b_{2,5,1}^{C,in}$	445.4634113411389
$t_{5,2}^{(C)}$	93.3	$b_{3,1,1}^{C,in}$	456.0
$t_{5,3}^{(C)}$	93.3	$b_{3,5,1}^{C,in}$	851.4065886588678
$t_{1,4,2}^H$	93.3	$b_{4,3,1}^{C,in}$	317.52
$t_{2,2,1}^H$	137.8	$b_{5,4,1}^{C,in}$	1459.7075544057818
$t_{2,5,1}^H$	198.85754716981077	$b_{1,4,2}^{C,out}$	1097.5220317460294
$t_{3,1,1}^H$	175.34864864864863	$b_{2,2,1}^{C,out}$	1352.37
$t_{3,5,1}^H$	158.19729729729815	$b_{2,5,1}^{C,out}$	975.9134113411451
$t_{4,3,1}^H$	148.9	$b_{3,1,1}^{C,out}$	1216.0
$t_{5,4,1}^H$	65.6	$b_{3,5,1}^{C,out}$	1865.2465886588552
$t_{1,4,2}^C$	129.3704699676358	$b_{4,3,1}^{C,out}$	1857.24
$t_{2,2,1}^C$	221.7	$b_{5,4,1}^{C,out}$	2507.597554405756
$t_{2,5,1}^C$	139.21155520203027		
$t_{3,1,1}^C$	160.0		
$t_{3,5,1}^C$	166.23812949640245		
$t_{4,3,1}^C$	221.1		
$t_{5,4,1}^C$	172.98757761639607		
$b_{1,4,2}^{H,in}$	1408.0		
$b_{2,2,1}^{H,in}$	1449.960117011749		
$b_{2,5,1}^{H,in}$	1188.3798829883126		
$b_{3,1,1}^{H,in}$	1069.4723774053136		
$b_{3,5,1}^{H,in}$	2285.687622594663		
$b_{4,3,1}^{H,in}$	3415.8599999999997		
$b_{5,4,1}^{H,in}$	1563.5808027006651		
$b_{1,4,2}^{H,out}$	821.0400000000004		
$b_{2,2,1}^{H,out}$	802.7501170117489		
$b_{2,5,1}^{H,out}$	657.9298829883065		
$b_{3,1,1}^{H,out}$	309.47237740531364		
$b_{3,5,1}^{H,out}$	1271.8476225946756		
$b_{4,3,1}^{H,out}$	1876.1399999999999		

3 *RecLMTD* Tangent Points

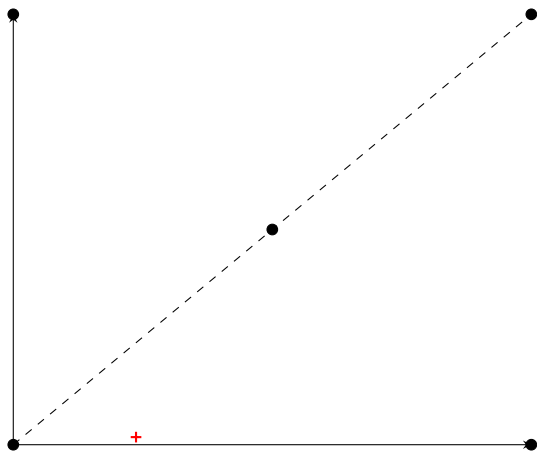
- (1, 4, 2)
 - Value: 0.037837
 - Error: 0.004098871066421
 - Relative error: 0.097740339029209
- (2, 2, 1)
 - Value: 0.040625
 - Error: 0.0000000000000001
 - Relative error: 0.0000000000000026
- (2, 5, 1)
 - Value: 0.028331
 - Error: 0.011259893016342
 - Relative error: 0.284403960693440
- (3, 1, 1)
 - Value: 0.015364
 - Error: 0.002432922323496
 - Relative error: 0.136704889390093
- (3, 5, 1)
 - Value: 0.051275
 - Error: 0.005670097583271
 - Relative error: 0.099571521877864
- (4, 3, 1)
 - Value: 0.013067
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (5, 4, 1)
 - Value: 0.043709
 - Error: 0.000002348739649
 - Relative error: 0.000053733528173
- $cu, 3$
 - Value: 0.018998
 - Error: 0.000000239104566
 - Relative error: 0.000012585722411
- $cu, 5$
 - Value: 0.015159
 - Error: 0.000042877764406
 - Relative error: 0.002820615997715



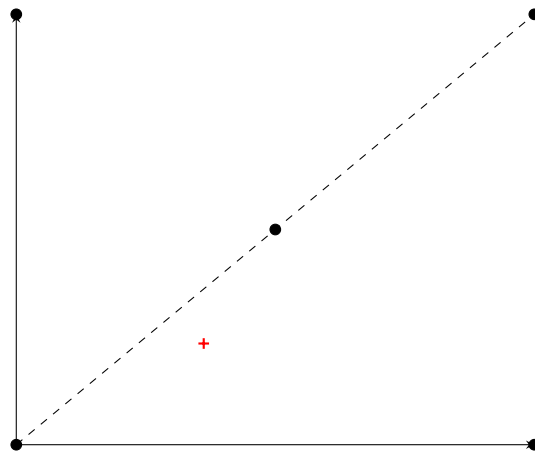
(a) RecLMTD - (1, 4, 2)



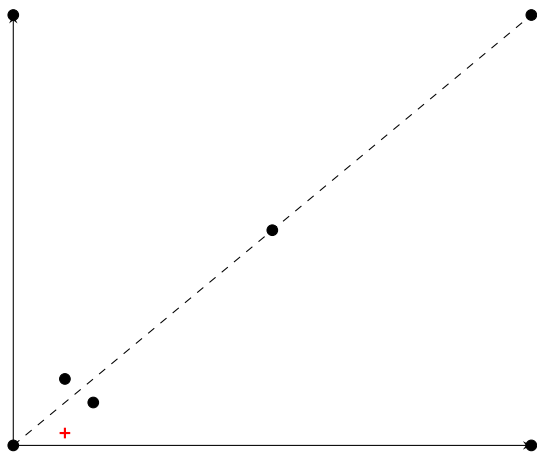
(b) RecLMTD - (2, 2, 1)



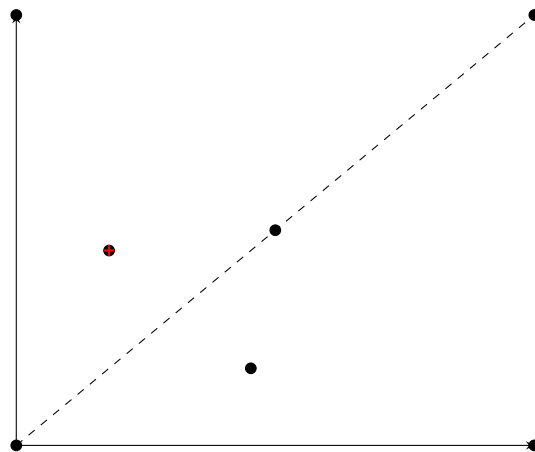
(c) RecLMTD - (2, 5, 1)



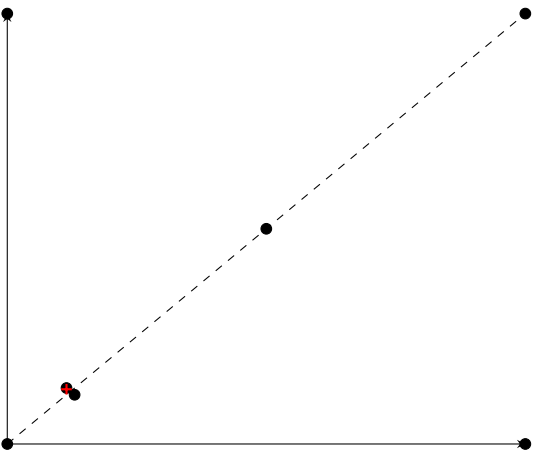
(d) RecLMTD - (3, 1, 1)



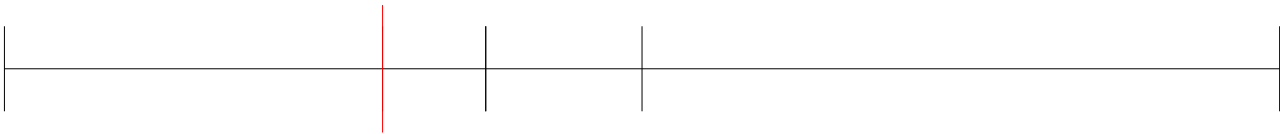
(e) RecLMTD - (3, 5, 1)



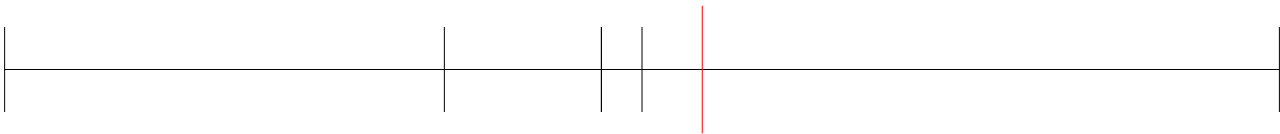
(f) RecLMTD - (4, 3, 1)



(g) *RecLMTD* - (5, 4, 1)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 4, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.960117011749
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 1)
 - Value: 1188.3798829883126
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 1, 1)
 - Value: 1069.4723774053136
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2285.687622594663
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 4, 1)
 - Value: 1563.5808027006651
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.454185642663975e-16

4.2 $b^{H,\text{out}}$

- (1, 4, 2)
 - Value: 821.0400000000004
 - Absolute error: 3.410605131648481e-13
 - Relative error: 4.1540060553060515e-16
- (2, 2, 1)

- Value: 802.7501170117489
 - Absolute error: 2.1259438653942198e-11
 - Relative error: 2.648325824365045e-14
- (2, 5, 1)
 - Value: 657.9298829883065
 - Absolute error: 291.52093515955437
 - Relative error: -0.3070416387951913
- (3, 1, 1)
 - Value: 309.47237740531364
 - Absolute error: 517.7465733890369
 - Relative error: -0.6258881918648772
- (3, 5, 1)
 - Value: 1271.8476225946756
 - Absolute error: 323.1660710990277
 - Relative error: -0.20261021731458975
- (4, 3, 1)
 - Value: 1876.1399999999999
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.2119227533298796e-16
- (5, 4, 1)
 - Value: 515.6908027006716
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -2.2045543012642127e-16

4.3 $b^{C,in}$

- (1, 4, 2)
 - Value: 510.5620317460293
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 1)
 - Value: 445.4634113411389
 - Absolute error: 5.684341886080802e-14
 - Relative error: -1.2760513526727547e-16

- (3, 1, 1)
 - Value: 456.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 851.4065886588678
 - Absolute error: 6.707523425575346e-12
 - Relative error: 7.878167158820161e-15
- (4, 3, 1)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 1459.7075544057818
 - Absolute error: 188.30093742163353
 - Relative error: -0.11425968880344398

4.4 $b^{C,out}$

- (1, 4, 2)
 - Value: 1097.5220317460294
 - Absolute error: 293.97397828356213
 - Relative error: 0.36584492615822545
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 1)
 - Value: 975.9134113411451
 - Absolute error: 311.24401928968007
 - Relative error: 0.46826892137915777
- (3, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 1865.2465886588552
 - Absolute error: 348.24510163246714

- Relative error: 0.22956147677553954
- (4, 3, 1)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 2507.597554405756
 - Absolute error: 52.68399145475814
 - Relative error: 0.021460629917832166

$$(a) \ t^{(H)} - (1, 2)$$

$$(a) \ t^{(H)} - (2, 1)$$

$$(a) \ t^{(H)} - (2, 1)$$

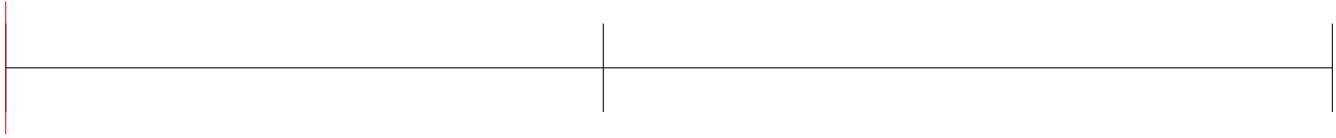
$$(a) \ t^{(H)} - (3, 1)$$

$$(a) \ t^{(H)} - (3, 1)$$

$$(a) \ t^{(H)} - (4, 1)$$

$$(a) \ t^{(H)} - (5, 1)$$

$$(a) \ t^{(H)} - (1, 4, 2)$$



$$(a) \ t^{(H)} - (2, 2, 1)$$



$$(a) \ t^{(H)} - (2, 5, 1)$$



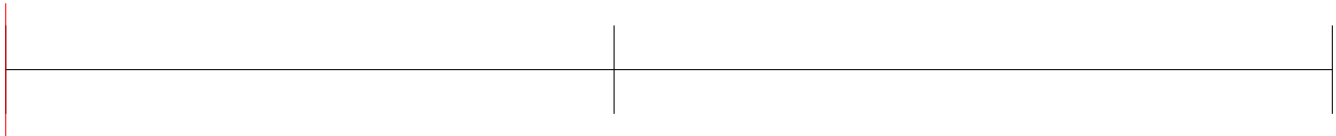
$$(a) \ t^{(H)} - (3, 1, 1)$$



$$(a) \ t^{(H)} - (3, 5, 1)$$



$$(a) \ t^{(H)} - (4, 3, 1)$$



$$(a) \ t^{(H)} - (5, 4, 1)$$



$$(a) \ t^{(C)} - (4, 3)$$



$$(a) \ t^{(C)} - (2, 2)$$



(a) $t^{(C)} - (5, 2)$



(a) $t^{(C)} - (1, 2)$



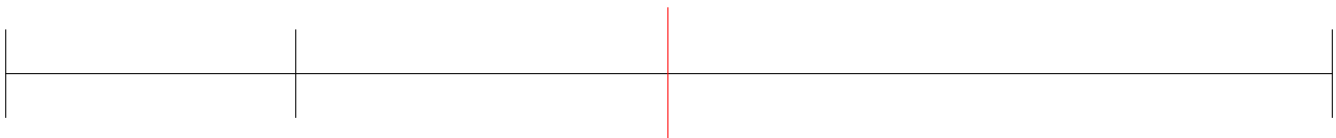
(a) $t^{(C)} - (5, 2)$



(a) $t^{(C)} - (3, 2)$



(a) $t^{(C)} - (4, 2)$



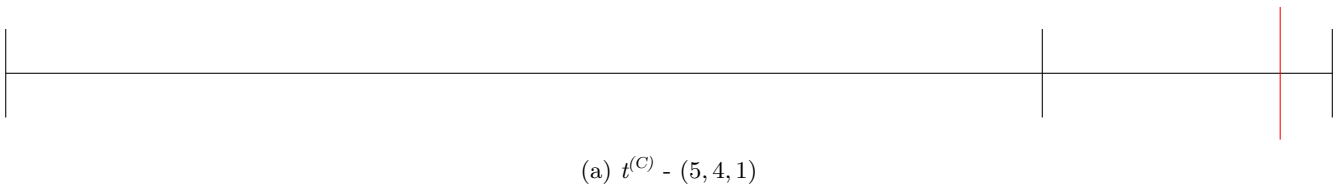
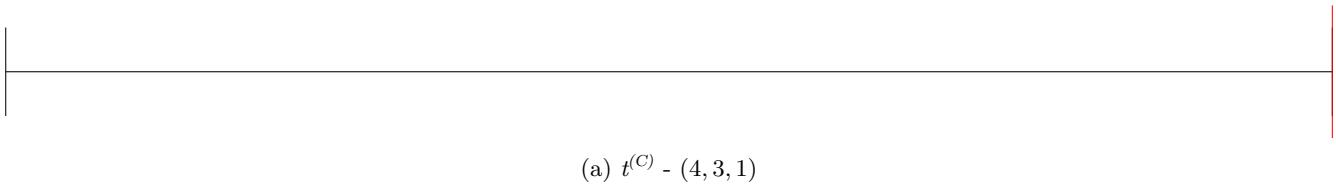
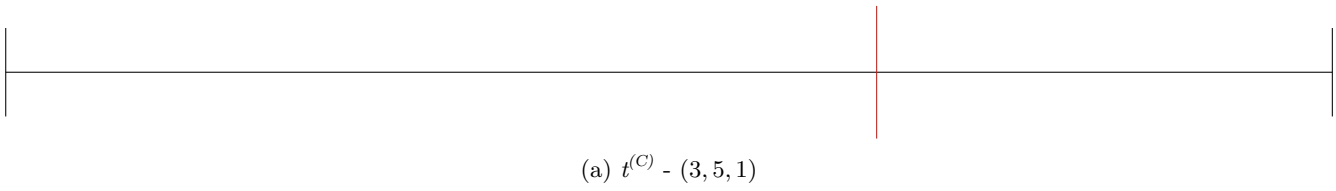
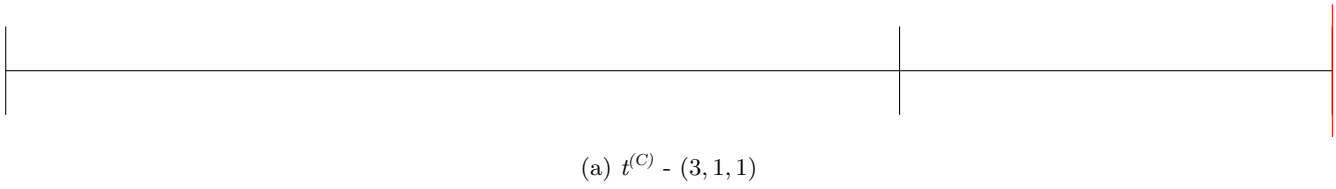
(a) $t^{(C)} - (1, 4, 2)$



(a) $t^{(C)} - (2, 2, 1)$



(a) $t^{(C)} - (2, 5, 1)$



5 Area Breakpoints

5.1 A

- $(1, 4, 2)$
 - Value: 26.128323
 - Error: 0.0000000000000082
 - Relative error: -0.000000000000003
- $(2, 2, 1)$
 - Value: 30.932783
 - Error: 0.0000000000000487
 - Relative error: 0.000000000000016
- $(2, 5, 1)$
 - Value: 4.010661
 - Error: 13.669738932839024
 - Relative error: 0.773157797574968
- $(3, 1, 1)$
 - Value: 13.737198
 - Error: 0.0000000000000364
 - Relative error: 0.000000000000027
- $(3, 5, 1)$
 - Value: 30.750914
 - Error: 30.407344184694171
 - Relative error: 0.497191143068009
- $(4, 3, 1)$
 - Value: 23.670440
 - Error: 0.0000000000000632
 - Relative error: 0.000000000000027
- $(5, 4, 1)$
 - Value: 38.682379
 - Error: 15.202008590564965
 - Relative error: 0.282122691575931

5.2 A_{CU}

- $cu, 3$
 - Value: 13.643579
 - Error: 0.0000000000000272
 - Relative error: -0.000000000000020
- $cu, 5$

- Value: 22.410514
- Error: 0.978798065301273
- Relative error: 0.041848090875162



(a) $A^\beta - (1, 4, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 5, 1)$



(a) $A^\beta - (3, 1, 1)$



(a) $A^\beta - (3, 5, 1)$



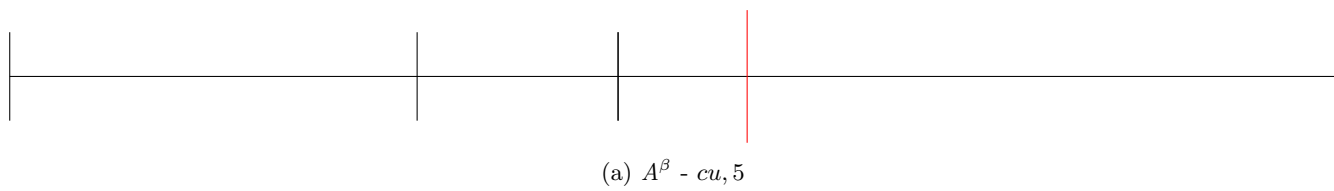
(a) $A^\beta - (4, 3, 1)$



(a) $A^\beta - (5, 4, 1)$



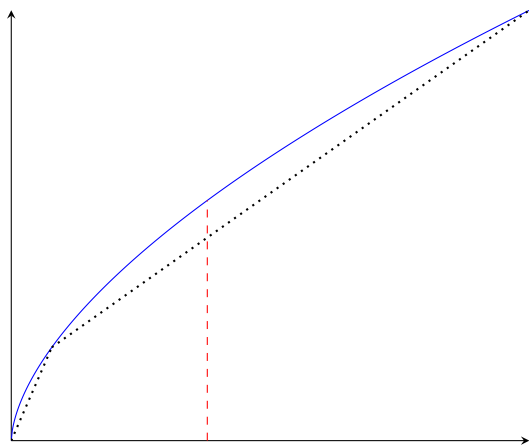
(a) $A^\beta - cu, 3$



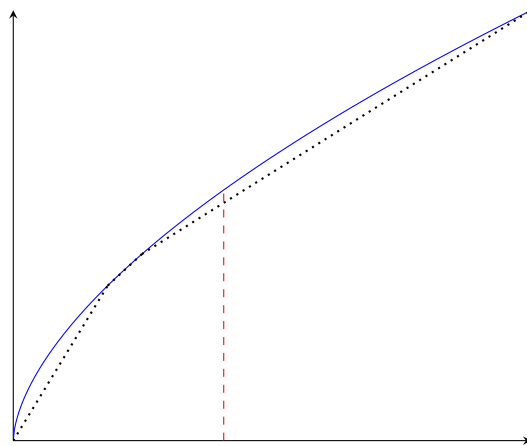
6 A^β Breakpoints

- (1, 4, 2)
 - q : 26.128323
 - $\hat{q}^\beta$: 5.988269
 - q^β : 7.083810
 - Breakpoints: [0, 5.40465749281718, 69.05411764705883]
 - Error: 1.095541707765970
 - Relative error: 0.154654297796964
- (2, 2, 1)
 - q : 30.932783
 - $\hat{q}^\beta$: 7.435268
 - q^β : 7.838833
 - Breakpoints: [0, 13.879417919656465, 18.890814020083003, 76.14235294117645]
 - Error: 0.403565024529700
 - Relative error: 0.051482795987421
- (2, 5, 1)
 - q : 4.010661
 - $\hat{q}^\beta$: 0.557925
 - q^β : 2.301069
 - Breakpoints: [0, 138.54823529411763]
 - Error: 1.743143941477111
 - Relative error: 0.757536720652917
- (3, 1, 1)
 - q : 13.737198
 - $\hat{q}^\beta$: 2.276881
 - q^β : 4.816581
 - Breakpoints: [0, 89.41176470588236]
 - Error: 2.539700492948070
 - Relative error: 0.527282815101646
- (3, 5, 1)
 - q : 30.750914
 - $\hat{q}^\beta$: 6.733274
 - q^β : 7.811147
 - Breakpoints: [0, 44.57376487924321, 181.68117647058827]
 - Error: 1.077872913191483
 - Relative error: 0.137991629445167
- (4, 3, 1)

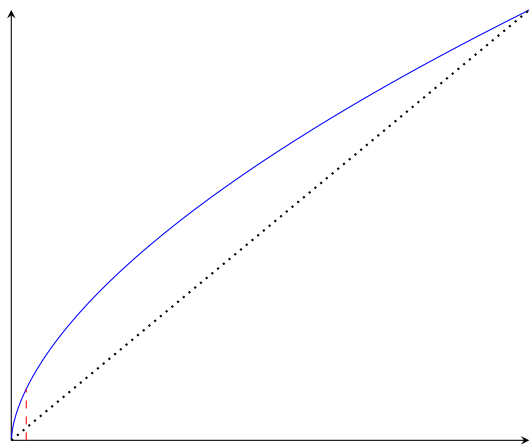
- q : 23.670440
- $\hat{q}^\beta$: 6.406341
- q^β : 6.676115
- Breakpoints: [0, 19.921388087224845, 181.14352941176475]
- Error: 0.269774269402269
- Relative error: 0.040408872132961
- (5, 4, 1)
 - q : 38.682379
 - $\hat{q}^\beta$: 7.341425
 - q^β : 8.964125
 - Breakpoints: [0, 12.878068451030797, 192.3352941176471]
 - Error: 1.622700441699736
 - Relative error: 0.181021619450815
- $cu, 3$
 - q : 13.643579
 - $\hat{q}^\beta$: 4.764278
 - q^β : 4.796859
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 16.738997089244645, 128.44338870924773]
 - Error: 0.032581128420707
 - Relative error: 0.006792179173195
- $cu, 5$
 - q : 22.410514
 - $\hat{q}^\beta$: 6.370670
 - q^β : 6.460574
 - Breakpoints: [0, 9.832103793393394, 14.682070969056184, 20.944873037188852, 127.1036185995295]
 - Error: 0.089903799132554
 - Relative error: 0.013915760690861



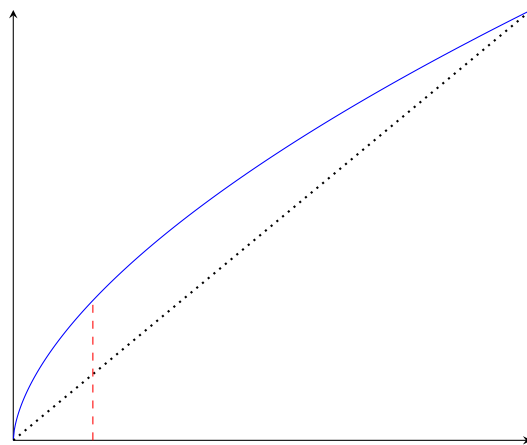
(a) $q^\beta - (1, 4, 2)$



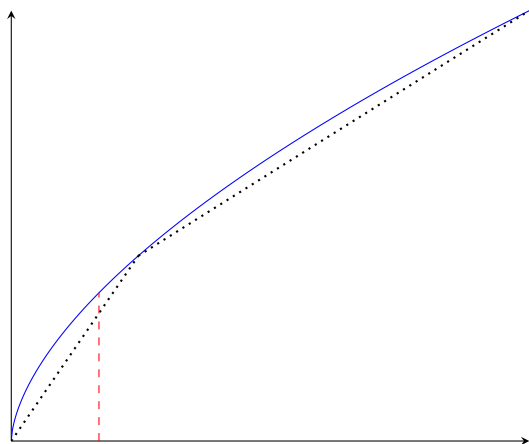
(b) $q^\beta - (2, 2, 1)$



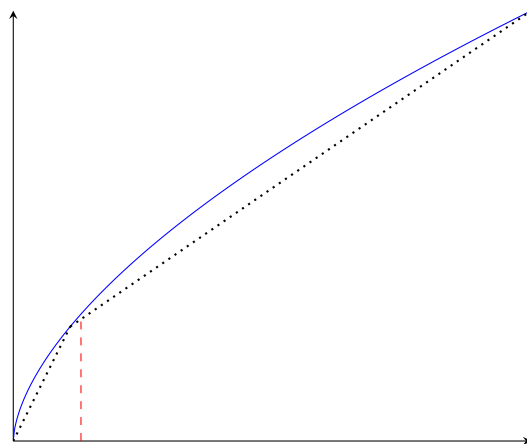
(c) $q^\beta - (2, 5, 1)$



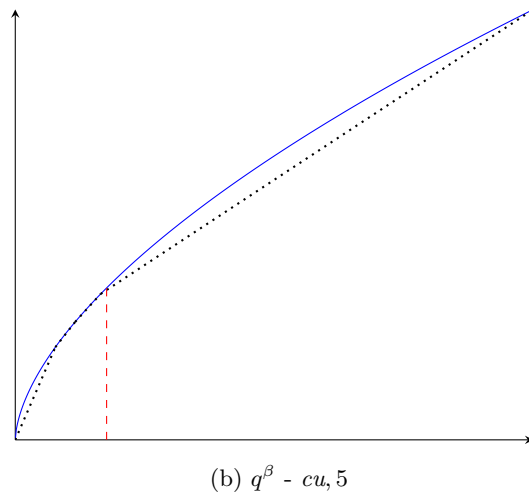
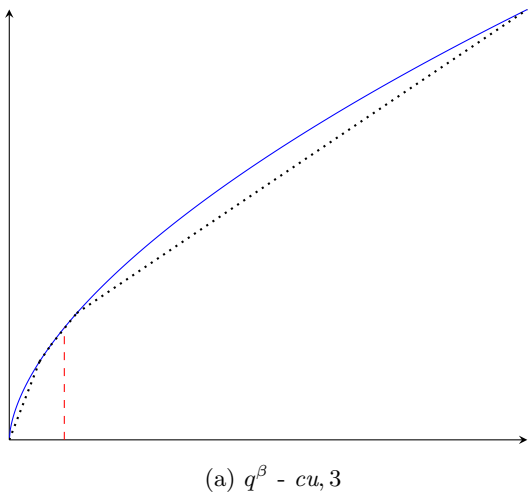
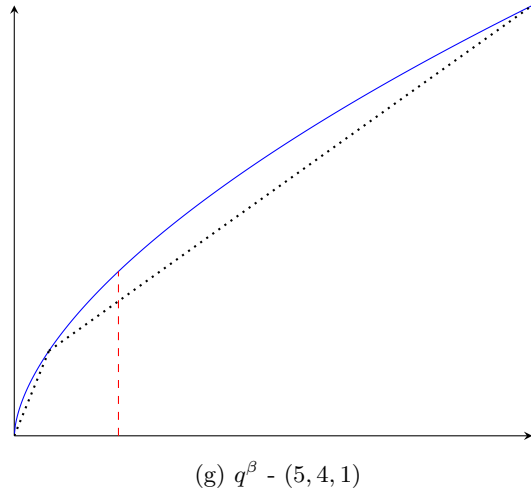
(d) $q^\beta - (3, 1, 1)$



(e) $q^\beta - (3, 5, 1)$



(f) $q^\beta - (4, 3, 1)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,4,2}^\beta$	0.0
$z_{1,1,2}$	0.0	$z_{5,1,1}$	-0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,2}$	-0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,2}$	-0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,2}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,2}$	-0.0	$A_{3,4,1}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,1}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{cu,1}$	-0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,2,2}$	-0.0	$z_{cu,2}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{hu,2}$	-0.0	$A_{4,3,2}^\beta$	0.0
$z_{2,4,2}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,1,2}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,2}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	-0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,3,2}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,5,1}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,1}$	111.1
$A_{hu,5}^\beta$	0.0	$A_{4,5,1}$	0.0	$dt_{2,1,2}$	10.0
$A_{1,1,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,1,3}$	100.0
$A_{1,1,2}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,2,1}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,2}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,2}$	183.3
$A_{1,3,1}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,2}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,4,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,4,2}$	21.671676300577957
$A_{1,5,1}$	0.0	$A_{5,4,2}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,1,2}$	0.0	$A_{cu,1}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,2,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,3,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,4,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,3,2}$	13.89308424131106
$A_{2,4,2}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,3}$	27.8
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,4,1}$	144.5
$A_{3,1,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,4,2}$	98.27167630057785
$A_{3,2,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,3}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,1,2}$	100.0	$dt_{3,5,3}$	10.0
$A_{3,3,1}$	0.0	$dt_{1,1,3}$	100.0	$dt_{4,1,1}$	122.2
$A_{3,3,2}$	0.0	$dt_{1,2,1}$	17.15006133296989	$dt_{4,1,2}$	26.24280137358479
$A_{3,4,1}$	0.0	$dt_{1,2,2}$	17.15006133296989	$dt_{4,1,3}$	100.0
$A_{3,4,2}$	0.0	$dt_{1,2,3}$	17.15006133296989	$dt_{4,2,1}$	122.2
$A_{3,5,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,2,2}$	106.1
$A_{4,1,1}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,2,3}$	106.1
$A_{4,1,2}$	0.0	$dt_{1,3,3}$	55.5	$dt_{4,3,3}$	183.3
$A_{4,2,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,4,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,4,2}$	10.0
$A_{4,3,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,3}$	94.5

Variable	Value	Variable	Value	Variable	Value
$dt_{4,5,1}$	122.2	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,5,2}$	10.0	$f_{2,1,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	111.1	$f_{2,2,2}^H$	0.0	$f_{1,1,2}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,3,1}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{5,1,2}$	10.0	$f_{2,3,2}^H$	10.6	$f_{1,2,2}^C$	0.0
$dt_{5,1,3}$	138.9	$f_{2,4,1}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,2,1}$	74.66004781348401	$f_{2,4,2}^H$	0.0	$f_{1,3,2}^C$	8.4
$dt_{5,2,2}$	10.0	$f_{2,5,2}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,3}$	74.66004781348397	$f_{3,1,2}^H$	0.0	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	0.0	$f_{1,5,2}^C$	13.9
$dt_{5,3,2}$	10.0	$f_{3,2,2}^H$	4.04450052976798	$f_{2,1,1}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,4,3}$	10.0	$f_{3,3,2}^H$	0.0	$f_{2,2,2}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,2}$	10.0	$f_{3,4,2}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,2}^H$	10.75549947023202	$f_{2,4,1}^C$	3.1087291771042103
$dt_{cu,1}$	53.3	$f_{4,1,1}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,2}^H$	0.0	$f_{2,5,2}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,2}^H$	11.260677840788803	$f_{3,2,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,3,2}^H$	1.3393221592111808	$f_{3,2,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,1,2}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{5,1,2}^H$	17.22935085853187	$f_{4,1,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,2,1}^H$	9.838859714928786	$f_{4,1,2}^C$	7.6
$f_{1,3,1}^H$	0.0	$f_{5,2,2}^H$	0.4706491414680891	$f_{4,2,1}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,2}^C$	6.1
$f_{1,4,1}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,3,2}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,2}^H$	0.0	$f_{4,4,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,2}^C$	11.0887830687831

Variable	Value	Variable	Value	Variable	Value
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.028945360806037865	$q_{1,3,1}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,2}$		$q_{1,3,2}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,5,2}$	0.08227027027027185	$q_{1,4,1}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.019647092310865477	$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.027922618193149045	$q_{2,1,1}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.023079286609400358	$q_{2,1,2}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,3,2}$	0.0064938037725891535	$q_{2,2,2}$	0.0
$f_{5,4,2}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$q_{2,3,1}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.024030191698075007	$q_{2,3,2}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,5,1}$	0.024784155348990776	$q_{2,4,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$q_{2,4,2}$	0.0
$\text{RecLMTD}_{1,1,2}$	0.1	$\text{RecLMTD}_{5,1,1}$		$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,2}$		$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,2,1}$		$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,2}$		$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.03070641897615819	$\text{RecLMTD}_{5,3,1}$		$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028797246571687217	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,2}$		$q_{3,4,1}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$		$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,1,1}$	0.1	$\text{RecLMTD}_{5,5,2}$		$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.023339478053072158	$\text{RecLMTD}_{cu,1}$	0.016108908592626668	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.024670522525974917	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.01505644655027976	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.016086786359643085	$\text{RecLMTD}_{hu,2}$	0.05607814773421494	$q_{4,3,2}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.1	$\text{RecLMTD}_{hu,3}$	0.06462248848511187	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.021215455068586544	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,5,1}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,1}$		$q_{4,5,2}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,1,2}$		$q_{5,1,1}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,1}$		$q_{5,1,2}$	0.0
$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,2,2}$		$q_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,1}^C$	129.7960526315789
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,1,2}^C$	60.0
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,2,2}^C$	115.6
$q_{5,4,2}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,1}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	65.6	$t_{2,3,2}^C$	37.8
$q_{5,5,2}$	0.0	$t_{3,5,2}^H$	122.35608108108104	$t_{2,4,1}^C$	117.26535667159649
$q_{cu,1}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,4,2}^C$	112.86184971098264
$q_{cu,2}$	0.0	$t_{4,1,2}^H$	148.9	$t_{2,5,2}^C$	131.46187050359833
$q_{cu,4}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,2}^C$	60.0
$q_{hu,1}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,2,1}^C$	115.6
$q_{hu,2}$	0.0	$t_{4,3,2}^H$	148.9	$t_{3,2,2}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	111.5
$q_{hu,4}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,3,2}^C$	37.8
$q_{hu,5}$	0.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	82.2
$t_{1,1,1}^H$	160.0	$t_{4,5,2}^H$	148.9	$t_{3,4,2}^C$	82.2
$t_{1,1,2}^H$	133.91477272727275	$t_{5,1,1}^H$	198.9	$t_{3,5,2}^C$	174.9066208547471
$t_{1,2,1}^H$	160.0	$t_{5,1,2}^H$	143.24166839309086	$t_{4,1,1}^C$	95.30426148707973
$t_{1,2,2}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,2}^C$	60.0
$t_{1,3,1}^H$	160.0	$t_{5,2,2}^H$	65.6	$t_{4,2,1}^C$	180.19804517446823
$t_{1,3,2}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,2}^C$	115.6
$t_{1,4,1}^H$	160.0	$t_{5,3,2}^H$	106.53559322033877	$t_{4,3,2}^C$	37.8
$t_{1,5,1}^H$	160.0	$t_{5,4,2}^H$	65.6	$t_{4,4,1}^C$	82.2
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,1,1}^H$	198.8575471698114	$t_{5,5,2}^H$	65.6	$t_{4,5,1}^C$	93.3
$t_{2,1,2}^H$	137.8	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,2}^C$	93.3
$t_{2,2,2}^H$	187.8424528301887	$t_{1,1,2}^C$	155.57058803725263	$t_{5,1,1}^C$	92.62135922330097
$t_{2,3,1}^H$	248.9	$t_{1,2,1}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,3,2}^H$	137.8	$t_{1,2,2}^C$	115.6	$t_{5,2,1}^C$	115.6
$t_{2,4,1}^H$	248.9	$t_{1,3,1}^C$	111.5	$t_{5,2,2}^C$	115.6
$t_{2,4,2}^H$	137.8	$t_{1,3,2}^C$	37.8	$t_{5,3,1}^C$	111.5
$t_{2,5,2}^H$	137.8	$t_{1,4,1}^C$	102.85953757225435	$t_{5,3,2}^C$	37.8
$t_{3,1,2}^H$	65.6	$t_{1,5,1}^C$	148.85	$t_{5,4,2}^C$	146.03815028901732
$t_{3,2,1}^H$	226.7	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	148.85

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	93.3	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,1,2}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	1316.0007652472045
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	2441.7654163197194	$b_{4,1,1}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	1956.9491972993355	$b_{4,1,2}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	30.87458368030412	$b_{4,2,1}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	1676.7149304934608
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,3,2}^{H,out}$	199.42506950654075
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,1,2}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	2441.7654163197194
$b_{2,3,2}^{H,in}$	1460.68	$b_{1,2,2}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	1956.9491972993355
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	30.87458368030412
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	0.0	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	265.31923475277944	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,1,2}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	1460.68	$b_{1,2,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	1316.0007652472045	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	317.52
$b_{4,1,2}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	0.0	$b_{1,5,1}^{C,in}$	0.0
$b_{4,2,2}^{H,in}$	1676.7149304934646	$b_{3,2,1}^{H,out}$	0.0	$b_{1,5,2}^{C,in}$	1296.87
$b_{4,3,2}^{H,in}$	199.42506950654337	$b_{3,2,2}^{H,out}$	265.31923475277944	$b_{2,1,1}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	0.0
$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,1}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	549.3124455942092	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	317.52	$b_{5,3,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	1296.87	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,1}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	549.3124455942334		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	456.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	705.16	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,2}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	911.4979682539708	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	456.0		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	705.16		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,3,2}^{C,out}$	0.0		
$b_{5,4,2}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	911.4979682539708		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0		
$b_{1,1,2}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0		